



Identical twins share all their genes, and, if raised together, much of their environment. Together, genes and environment determine many aspects of behavior.

gave the field its name was published in 1960, and a decade later the Behavior Genetics Association was founded. For a time, most behavior genetics research was an effort to show that the term was not itself an oxymoron—that variations in genes do contribute to individual differences in behavior. Now, as a result of that research, the relevance and importance of genetic variation to individual differences in behavior are widely accepted, and the challenging task is to identify specific gene-behavior pathways. In this entry, we will review the methods used to identify such pathways and then focus on one set of behaviors, use and abuse of alcohol, as a model for the study of genetic and environmental influences.

Twin and Adoption Studies

To determine whether variation in some dimension of behavior is heritable (whether behavioral differences are, in some part, due to genetic differences between people), human researchers use family, twin, and adoption designs. The first step in determining whether a behavior is influenced by genes is to establish that it aggregates or “runs” in families. Similarities in behavioral characteristics among family members suggest that genes influence the trait, but they cannot conclusively demonstrate genetic influence, because family members share their experiences (i.e., their environments) as well as their genes.

Twin and adoption studies allow one to tease apart the effects of genes and environments. Twin studies compare the patterns of behavioral characteristics between identical, or monozygotic (MZ), and fraternal, or dizygotic (DZ), twins. MZ twins share 100 percent of their genetic information, whereas DZ twins share, on average, one-half, just like non-twin siblings. Thus, the presence of greater behavioral similarities between MZ twins than DZ twins suggests that genetic factors contribute to those behaviors.

Adoption studies compare whether an adopted child is more similar behaviorally to the child's adoptive parents (with whom environments, but not genes, are shared) or to the child's biological parents (with whom genes, but not environments, are shared). Twin and adoption techniques have been used to demonstrate that nearly all behavior is under some degree of genetic influence, and, in the context of the Human Genome Project, behavior genetics has attracted great interest and some controversy.

Complex Genetics

New techniques allow behavior geneticists to ask not just whether a behavior is under genetic influence, but also what specific genes are involved. To identify genes involved in behavior, investigators use genetic markers—stretches of DNA that differ among individuals. One can either use genetic markers that are evenly spaced on all chromosomes, to search for genes influencing the behavior that are located anywhere in the genome (called genomic screening), or one can test markers at a specific gene believed to be, on theoretical grounds, involved in the behavior (called the candidate gene approach).

The idea behind these analyses is that if a particular gene is involved in the behavior, then people who are more alike with respect to the behavior will be more likely to share the same stretch of DNA that is at or near the gene. The difficulty in searching for genes involved in behavior is that there is no one-to-one correspondence between carrying a particular gene and exhibiting a particular behavior. There are no genes *for* behavior; there are only genes that *influence* behavior. Any particular behavior is a complex trait that involves more than one gene and is influenced by the environment as well.

For example, having a particular gene may make a person *more likely* to have problems with alcohol, but it does not *determine* whether or not the person will be an alcoholic. Some individuals will carry genes predisposing them to alcohol abuse but will never exhibit any problems, because they choose to abstain from alcohol. Other individuals will exhibit obvious alcohol problems, but will not carry the particular genes known to be involved.

This is because a large number of genes are risk-relevant for use and abuse of alcohol, and each has only a very small effect. Different genes may be acting in different individuals. And genes interact with each other and with the environment. Thus, individual outcomes result from a complex and ill-understood mixture of both genetic and environmental risk factors. That very complexity creates the diverse nature of human behavior. Indeed, it is what makes us uniquely human, but it also makes finding genes involved in human behavior extraordinarily difficult.

Animal Models

Because of this complexity, some investigators use animal models to complement human studies. Like humans, mice and rats differ in a variety of behavioral characteristics, including levels of alcohol use and tolerance or sensitivity to its effects. Animal studies allow breeding strategies that cannot be performed in humans. One approach that is commonly used in animal studies takes advantage of natural variation in behavior. Different

strains of mice differ not only in coat color but also in preference for alcohol.

Under one of the most commonly used breeding strategies, animals from each of the behaviorally different mouse lines are allowed to mate with each other. Assuming the parents from each strain have different versions of genes contributing to alcohol use, subsequent generations of offspring will have different combinations of the genes contributing to the alcohol use and will display wide variation in their alcohol use. Such samples can be used to perform genetic studies searching for genes involved in the behavior, much like those described in humans: Animals more alike in their drinking behavior should be more likely to have inherited common stretches of DNA involved in the behavior.

One advantage of using animals is that the factors contributing to alcohol use in mice and rats are thought to be much simpler than the processes contributing to abuse in humans. Another is that animals' experience with alcohol can be experimentally controlled. Other strategies that are used in animals include inducing mutations or **"knocking out"** particular genes and studying the resultant aberrant behavior. If altering a particular gene consistently causes an alteration in a given behavior, the gene is likely involved in that behavior.

knocking out deleting a gene or obstructing gene expression

Alcoholism in Humans

The techniques available for human research are more limited, and many questions remain. Although behavior geneticists now possess the techniques to identify genetic influence and to begin to identify specific genes, questions remain regarding which behaviors, actions, and activities of people are the best candidates for behavior-genetic study.

Again, alcohol use and abuse provide an illustration. Alcoholism is a major social and medical problem in the United States and in most of the world. It is estimated that 10 percent of men and 4 percent of women in the United States experience alcohol dependency, at a cost of billions of dollars and 100,000 lives annually. Because use of alcohol is typically part of social interactions, familial (and possibly genetic) factors would be expected to contribute to variation in drinking.

But where shall we begin its study? Perhaps with diagnosed alcoholism? Most adults in our society use alcohol, yet only a fraction of them ever experience clinical symptoms of alcoholism. Perhaps we should begin much earlier, studying the decision to begin drinking? Obviously, one cannot become alcoholic without initiating drinking and then drinking large quantities regularly and with high frequency. Or perhaps much earlier yet, for behavioral predictors of alcoholism can be identified years before alcohol is first consumed.

Such predictors are apparent in early childhood, in behaviors evident to the children's parents, teachers, and peers. Long-term (i.e., longitudinal) studies conducted in several countries suggest that, as early as kindergarten and elementary school, behavioral ratings made by parents, teachers, or classmates distinguish children who are more likely to abuse alcohol later, in adolescence and early adulthood. Children who were impulsive, exploratory, excitable, curious, and distractible—and those who were less



cautious, less fearful, less shy, and less inhibited—have a much greater risk of adult alcoholism than do children without those characteristics.

Twin studies have demonstrated that additive genetic variance, as well as familial-environmental influences, significantly contributes to the childhood behaviors that play a central role in the development of alcoholism risk. So, to understand the development of alcoholism, one must appreciate the complex developmental influences that affect children years before they first consume alcohol. Those influences reflect the interactions of dispositional differences in children's behavior with variations in their familial, social, and school environments.

Twin Studies of Alcoholism

That risk-related behaviors are evident early in life, remain stable into adolescence, and are associated with a family history of alcoholism suggests that those behaviors are, at least in part, of genetic origin. To establish that, researchers must use genetically informative study designs.

One approach is to study child or adolescent twins and their parents. Several such studies, which specifically assess the initiation of alcohol use and the transition to alcohol abuse, are being conducted throughout the world. We illustrate with two ongoing studies from Finland.

One, "FinnTwin12," is a study of approximately 2,800 twin pairs and their parents. The twins represent all pairs from five consecutive twin-birth cohorts (1983–1987) who were entered into the study as they reached age twelve (1995–1999), when behavioral ratings by teachers and parents were obtained on all participating pairs.

The ratings include multidimensional scales (i.e., scales that rate various characteristics) of behaviors associated with increased alcoholism risk. Two years later, at age fourteen, the twins were followed up, and, while most reported abstinence, about one-third were then using alcohol.

What predicts drinking or abstaining at age fourteen? Genetic factors played a role only among twin sisters, perhaps reflecting their more accelerated pubertal maturation, and environmental effects shared by twin siblings accounted for most of the variation in drinking or abstaining at this age. Differences that twins attributed to their home environments (e.g., in parental monitoring, support, and understanding) and differences in teachers' ratings of twins' behavior at age twelve (in problem behaviors of aggressiveness, impulsivity, and inattention) differentiated those who were drinking from those still abstaining at fourteen.

But once drinking is initiated, genetic effects become evident in individual differences in frequency and quantity of consumption and in behavioral problems that then result. "FinnTwin16," another study of five consecutive, complete birth cohorts of Finnish twins, illustrates. These twins were first studied as they reached age 16, with follow-up twelve and thirty months later, at ages 17 and 18½. At age 16, about 25 percent had remained abstinent.

Of 2,810 twin pairs, both twins in 459 pairs (16.3%) were abstaining, co-twins in 1,964 pairs (69.9%) had concordantly begun drinking by age sixteen, and only 387 pairs were discordant, with one twin drinking and the

other abstaining. Concordance is the co-occurrence of the behavior in the twin pair (e.g., both drinking or both not drinking). Overall concordance exceeded 85 percent, regardless of the twins' gender or zygosity.

There was extremely high familial aggregation for alcohol use or abstinence at age sixteen, additional evidence that genes play little role in abstinence or initiation. But thirty months later, individual abstinence had dropped to 10 percent, concordance among twin pairs had declined considerably, and genetic factors increasingly influenced the frequency and quantity of an adolescent's alcohol consumption. MZ twins were significantly more similar in drinking frequency than were DZ twins. The influence of genetic factors increases over time, with increasing experience with alcohol, and the differences between MZ and DZ twins becomes greater at each follow-up.

Regional residency moderates parental and sibling influences on adolescent drinking. Where abstinence is relatively rare, as in the large cities of Finland, siblings have greater effects on one another. Conversely, the protective effect of parental abstinence on that of their adolescent twin children was more evident in sparsely populated rural areas of the country, where abstinence was more prevalent. And, most interestingly, genetic factors exerted a larger role in urban settings than in rural settings from age 16 through the follow-up at age 18½. Common environmental factors assumed greater importance in rural settings.

Such results suggest that environments moderate the impact of genetic effects across many dimensions of behavior. But what aspects of the environment matter? In an analysis of results at age 18½, we demonstrated that specific characteristics of rural and urban environments moderate the effects of genes on drinking behavior. In areas with proportionately more young adults, genetic effects were nearly five times more evident than in communities with relatively few young adults. Thus, dramatic differences in the magnitude of genetic effects can be demonstrated across communities at environmental extremes of specific risk-relevant characteristics.

Complex Behaviors, Complex Causes

Thus, for use and abuse of alcohol, we know that the importance of genetic and environmental effects changes with sequencing in the use and abuse of alcohol, from abstinence or initiation to frequency of regular consumption, to problems associated with consumption, and ultimately, to diagnosed alcoholism and end-organ damage from the cumulative effects of alcohol. Similar stories could be told for many other behaviors of interest. Thus, for the major psychopathologies, from depression and schizophrenia in adults to attention deficit disorder in children or eating disorders in adolescents, genetic influences are invariably part of the story but never the whole story.

Genetic effects are always probabilistic and not deterministic. And the action of genes on behavioral outcomes is likely to be indirect. So we conclude with the same message with which we began: There are no genes *for* behavior, but behavioral development always represents an exquisite interplay between genes and environments. Gene-behavior correlations are modest and nonspecific; they alter risk but rarely determine outcome. Genes represent dispositions, not destinies.

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Bioinformatics

Bioinformatics is the use of mathematical, statistical and computer methods to analyze biological, biochemical, and biophysical data. Because bioinformatics is a young, rapidly evolving field, however, it also has a number of other credible definitions. It can also be defined as the science and technology of learning, managing, and processing biological information. Bioinformatics is often focused on obtaining biologically oriented data, organizing this information into databases, developing methods to get useful information from such databases, and devising methods to integrate related data from disparate sources. The computer databases and **algorithms** are developed to speed up and enhance biological research.

Bioinformatics can help answer such questions as whether a newly analyzed gene is similar to any previously known gene, whether a protein's sequence can suggest how the protein functions, and whether the genes turned on in a cancer cell are different from those turned on in a healthy cell.

Databases and Analysis Programs

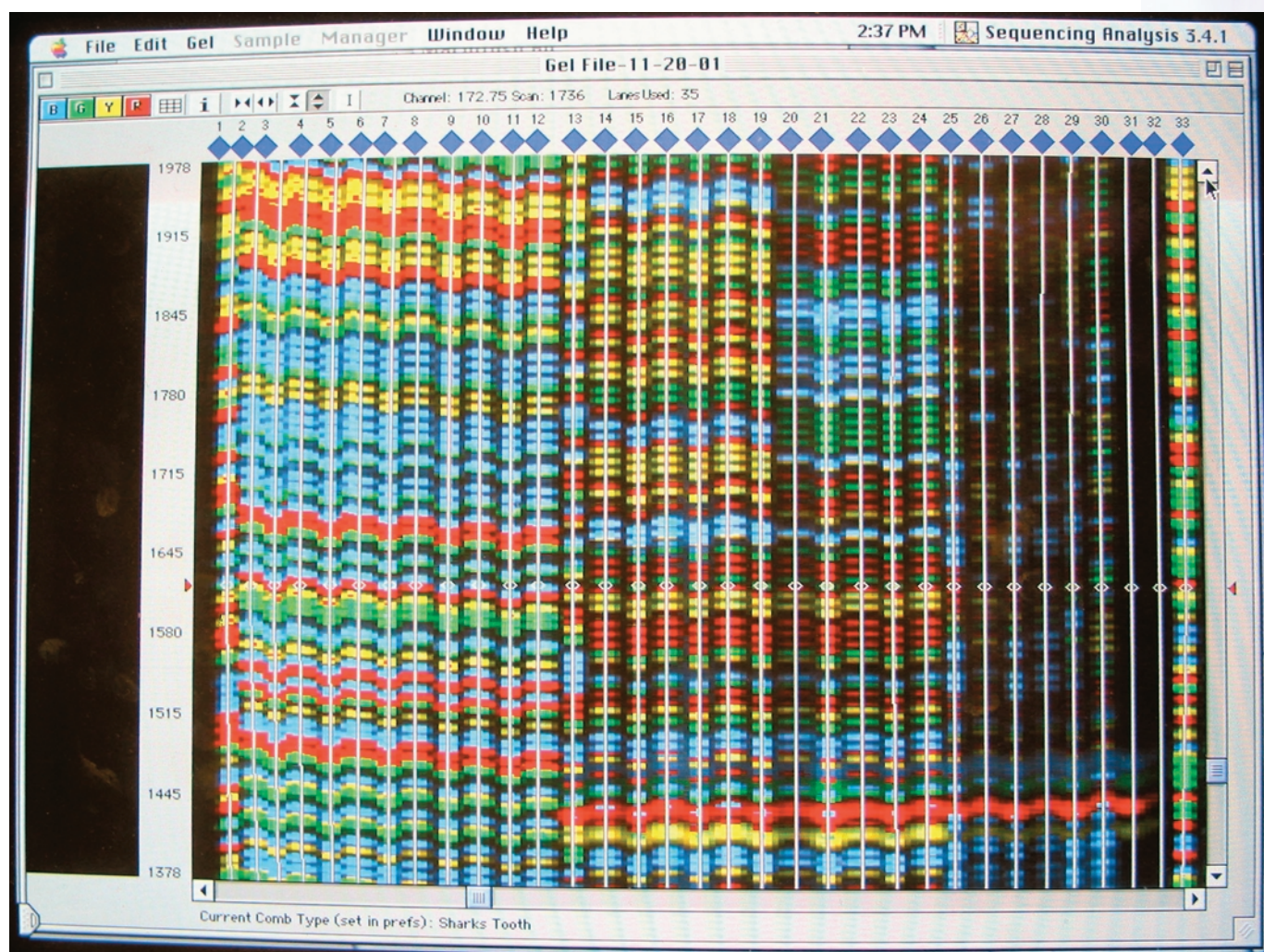
A good deal of the early work in bioinformatics focused on processing and analyzing gene and protein sequences catalogued in databases such as GenBank, EMBL, and SWISS-PROT. Such databases were developed in academia or by government-sponsored groups and served as repositories where scientists could store and share their sequence data with other researchers. With the start of the Human Genome Project in 1990, efforts in bioinformatics intensified, rising to the challenge of handling the large amounts of DNA sequence data being generated at an unprecedented rate. By the mid-to late-1990s, much of the efforts in bioinformatics centered around genomic data, generated by the Human Genome Project and by private companies, and around **proteomic** data.

Early analysis of sequence information focused on looking for similarities between genes and between proteins. Algorithms were developed to help researchers rapidly identify similar gene or protein sequences. Such tools were extremely useful for determining whether a newly sequenced piece of DNA was at all similar to sequences already entered in a database. To determine how multiple sequences align and to view their similarities, multiple-alignment programs were developed. Such programs helped scientists compare the sequences of closely related genes or compare the sequence of a particular gene or protein as it appears in several species.

To better understand the functional roles of new nucleotide and amino acid sequences, researchers developed algorithms to look for particular sequence "domains." Domains are regions where a particular sequence of

algorithms procedures or sets of steps

proteomic derived from the study of the full range of proteins expressed by a living cell



nucleotides or amino acids is indicative of function in the protein. For example, a protein may have a domain that binds to ATP or GTP, two important protein regulators.

In addition, these algorithms can detect sequences that denote a region involved in particular types of post-translational modifications, such as tyrosine **phosphorylation**. Tools such as prosite, blocks, prints, and Pfam can be used to detect and predict such protein domains in sequence data.

Structure is central to protein function, and another set of tools, including SWISS-MODEL, allows researchers to use gene and protein sequence data to predict a protein's three-dimensional structure. Such tools can help predict how mutations in a gene sequence could alter the three-dimensional structure of the corresponding protein. They accomplish such molecular modeling by comparing a novel sequence to the sequences of genes whose protein structures are known.

The majority of tools were developed as academic freeware distributed on the Internet. In the early- to mid-1990s, commercial companies began to develop their own **proprietary** algorithms and tools, as well as their own proprietary databases. Those databases were then marketed to pharmaceutical

The computer monitor of an automated gel sequencer displays a digital gel image. This data is more easily analyzed in this environment.

phosphorylation addition of the phosphate group PO_4^{3-}

proprietary exclusively owned; private

and biotech companies as well as to academic research groups. The most commercially viable and profitable businesses focused on the production and sale of proprietary DNA- and gene-sequence databases in the mid- to late-1990s. These databases primarily contained genetic information that were not in the public domain databases, such as GenBank, and they thus offered potential competitive advantages to the drug discovery groups of large pharmaceutical and biotech companies.

Applications of Bioinformatics to Drug Discovery

The application of bioinformatics to genomics data could be a huge potential boon for the discovery of new drugs. During the 1990s many pharmaceutical companies and biotech companies became convinced that they could speed up their drug-discovery pipelines by taking advantage of the data from the Human Genome Project as well as by funding their own internal genomics programs and by collaborating with third-party genomics companies.

The goal in such practical applications is to use such data as DNA sequence information and gene expression levels to help discover new drug targets. The vast majority of drugs target proteins, but there are a handful of drugs, such as some **chemotherapeutic** agents, that bind to DNA. In cases where the target is a protein, the drugs themselves are primarily small chemical molecules or, in some cases, small proteins, such as hormones, that bind to a larger protein in the body. Some drugs are therapeutic proteins delivered to the site of the disease.

The extent to which genomics will actually be able to help identify validated drug targets is uncertain. Genomics and bioinformatics are still young areas, and the drug development cycle can take up to ten years. As of 2001 relatively few of the drugs on the market or in the late stages of clinical trials were discovered via genomics or bioinformatics programs.

Specialists

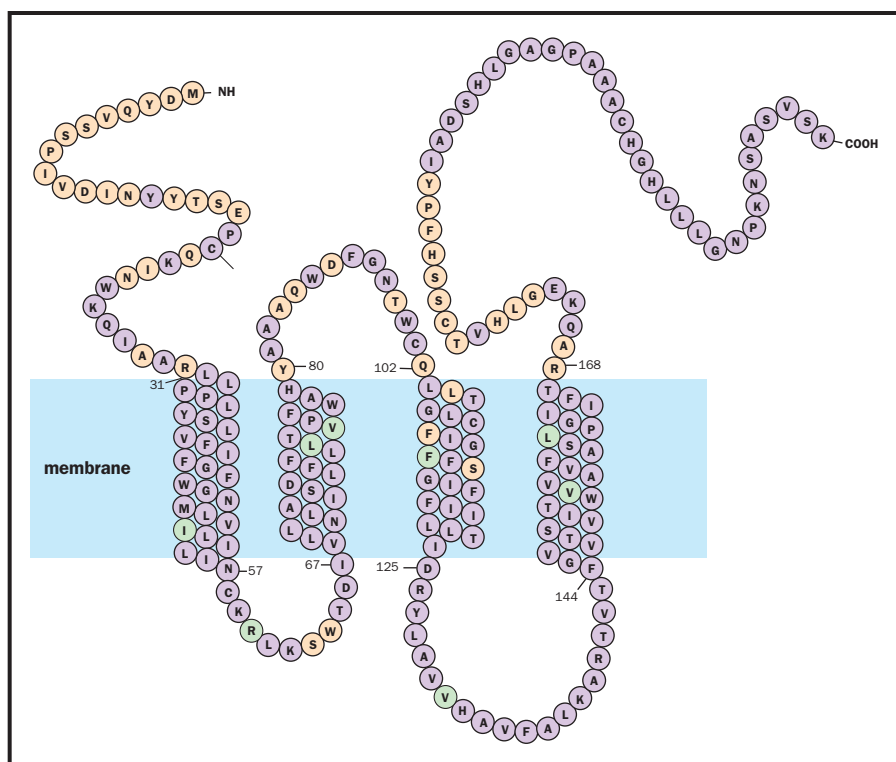
Bioinformatics is applied to at least five major types of activities: data acquisition, database development, data analysis, data integration, and analysis of integrated data.

Data Acquisition. Data acquisition is primarily concerned with accessing and storing data generated directly off of laboratory instruments. Many of these instruments are either automated or semi-automated **high-throughput** instruments that generate large volumes of data. The Human Genome Project utilized hundreds of DNA sequencers, producing enormous amounts of data. The data had to be captured in the appropriate format, and it had to be capable of being linked to all the information related to the DNA samples, such as the species, tissue type, and quality parameters used in the experiments. This area of bioinformatics primarily relates to the use of “laboratory information management systems,” which are the computer systems used to manage the information needs of a particular laboratory.

Database Development. Many laboratories generate large volumes of such data as DNA sequences, gene expression information, three-dimensional molecular structure, and high-throughput screening. Consequently, they must develop effective databases for storing and quickly accessing data. For each type of data, it is likely that a different database organization must be

chemotherapeutic use of chemicals to kill cancer cells

high-throughput rapid, with the capacity to analyze many samples in a short time



Predicted structure of a chemokine receptor in the membrane of a white blood cell. Knowing the amino acid sequence (represented by single letters) allows researchers to predict the three-dimensional structure of a protein. Adapted from <http://www.cdc.gov/ncidod/eid/vol3no3/smith.htm>.

used. A database must be designed to allow efficient storage, search, and analysis of the data it contains. Designing a high-quality database is complicated by the fact that there are several formats for many types of data and a wide variety of ways in which scientists may want to use the data. Many of these databases are best built using a relational database architecture, often based on Oracle or Sybase.

A strong background in relational databases is a fundamental requirement for working in database development. Having some background in the molecular biology techniques used to generate the data is also important. Most critical for the bioinformatics specialist is to have a strong working relationship with the researchers who will be using the database and the ability to understand and interpret their needs into functional database capabilities.

Data Analysis. Being able to analyze data efficiently requires having a good database design, allowing researchers to query the database effectively and letting them quickly obtain the types of information they need to begin their data analysis. If queries cannot be performed, or if performance is tediously slow, the whole system breaks down, since scientists will not be inclined to use the database. Once data is obtained from the database, the user must be able to easily transform it into the format appropriate for the desired analysis tools.

This can be challenging, since researchers often use a combination of publicly available tools, tools developed in-house, and third-party commercial tools. Each tool may have different input and output formats. Starting in the late 1990s, there have been both commercial and in-house efforts at pharmaceutical and biotech companies to reduce the formatting complexities.



gene expression use of a gene to create the corresponding protein

transduction conversion of a signal of one type into another type

Such simplification efforts focus on building analysis systems with a number of tools integrated within them such that the transfer of data between tools appears seamless to the end user.

Bioinformatics analysts have a broad range of opportunities. They may write specific algorithms to analyze data, or they may be expert users of analysis tools, helping scientists understand how the tools analyze the data and how to interpret results. A knowledge of various programming languages, such as Java, PERL, C, C++, and Visual Basic, is very useful, if not required, for those working in this area.

Data Integration. Once information has been analyzed, a researcher often needs to associate or integrate it with related data from other databases. For example, a scientist may run a series of **gene expression** analysis experiments and observe that a particular set of 100 genes is more highly expressed in cancerous lung tissue than in normal lung tissue. The scientist might wonder which of the genes is most likely to be truly related to the disease. To answer the question, the researcher might try to find out more information about those 100 genes, including any associated gene sequence, protein, enzyme, disease, metabolic pathways, or signal **transduction** pathway data.

Such information will help the researcher narrow the list down to a smaller set of genes. Finding this information, however, requires connections or links between the different databases and a good way to present and store the information. An understanding of database architectures and the relationship between the various biological concepts in the databases is key to doing effective data integration.

Analysis of Integrated Data. Once various types of data are integrated, users need a good way to present these various pieces of data so they can be interpreted and analyzed. The information should be capable of being stored and retrieved so that, over time, various pieces of information can be combined to form a “knowledge base” that can be extended as more experiments are run and additional data are integrated from other sources. This type of work requires skills related to database design and architecture. It also requires specific programming skills in various computer languages, as well as expertise in developing interfaces between a computer and its user. SEE ALSO COMBINATORIAL CHEMISTRY; COMPUTATIONAL BIOLOGIST; EVOLUTION OF GENES; GENOMICS; GENOMICS INDUSTRY; HIGH-THROUGHPUT SCREENING; HUMAN GENOME PROJECT; PHARMACOGENETICS AND PHARMACOGENOMICS; PROTEINS; PROTEOMICS; SEQUENCING DNA.

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Plants, growing in the wild or in cultivation, face numerous threats from insects, bacteria, viruses, and fungi, as well as from other plants. Biopesticides are inert substances or living organisms that can help protect plants from such threats. Chemical pesticides can offer similar protection but, by contrast, are neither alive nor made by living organisms.

A variety of chemicals produced by plants help ensure that parasites, predators, plant feeders, and **herbivores** seldom increase in number sufficiently to destroy the plant populations they prey upon. Chemicals found in very low concentrations in certain plants have been found to help keep locusts from feeding on those plants, and some trees produce nearly 1,000 different chemical compounds that help them resist herbivores and parasites.

nematodes worms of the Nematoda phylum, many of which are parasitic

Plant predators are themselves subject to attack by predators, parasites, and microbes, all of which can indirectly help protect a plant and therefore are also considered biopesticides. An oak tree may have about 100 species of insect herbivores feeding on it. In turn, there can be up to 1,000 species of predators, parasites, and microbes feeding on the herbivores. The microbes, parasites, and predators attacking the herbivore populations are considered “biopesticides,” as are any protective chemicals produced by the tree.

Such living biopesticides play a vital role in agriculture and nature, helping to control insect pests, plant pathogens, and weeds. Numerous organisms, including viruses, fungi, protozoa, bacteria, and **nematodes**, as well as insects, such as parasitic wasps, can attack pest insects and weeds. In some cases, biologists search around the world to find natural organisms to help control an insect, a plant **pathogen**, or weed populations.

The use of natural organisms as biopesticides is sometimes hampered by the presence of chemical pesticides, which can threaten populations of a pest insect's natural enemies. Pest outbreaks that result from chemical pesticides destroying a pest's natural enemies are estimated to cost the United States more than \$500 million per year.

Since the 1980s, many crops have been genetically modified to produce biopesticides that will help protect them from insects and pathogens (including viruses). In 1998, 40 million hectares of engineered crops were planted throughout the world (though 74% of the modified cropland is in the United States). Globally, 20 percent of this area has been planted with herbicide-tolerant crops, 8 percent with insect-resistant crops, and 0.3 percent with insect- and herbicide-resistant crops.

More than 95 percent of all crops have some degree of pathogen resistance bred into them, with resistance to fungi, bacteria, and viruses being most common. Most of this resistance was either added by farmer selection or

plant breeder selection, rather than through genetic engineering. It is because of this natural resistance that has been bred into the crops that only 12 percent of the pesticides used in U.S. agriculture are fungicides.

Some viral resistance, however, has been bred into a number of crops through insertion of viral genes into the plant chromosomes. These genes may lead to the plant's producing viral proteins—biopesticides of a sort—that hamper a virus's own actions. This pathogen-derived resistance has been successfully used to protect Hawaii's papaya crop from the devastating papaya ringspot potyvirus. The viral gene was inserted into the papaya genome using a "gene gun," which shoots viral genes into papaya embryo cells.

Insect-Resistant Crops

About 8 percent of land covered by genetically modified crops is planted with insect-resistant crops. Although insect-resistant crops have not been employed as widely as disease-resistant crops, there are some notable examples. These include Hessian fly resistance in wheat and European corn borer resistance in corn. Resistance to corn borers has been provided by using a naturally occurring toxin produced by *Bacillus thuringiensis* (BT). This bacterium has traditionally been applied to corn and other crops. Genetic engineering has allowed the toxin to be manufactured by the corn plant itself.

The most serious problem with the use of BT toxin genes has been with STARLINK corn. This variety of genetically modified corn was approved for use only as animal feed, not as human food. STARLINK corn nevertheless found its way into the food-processing industry when farmers did not keep their STARLINK corn separate from corn to be used as human food. Some grain elevator operators as well did not keep the STARLINK corn separated. The STARLINK mix-up cost the United States an estimated \$5 billion of processed food that had to be destroyed because of the BT toxin restriction.

Plants that have been genetically modified to make BT toxin produce it in every cell. BT corn pollen may harm nontarget moths and butterflies, like Monarch butterflies. Milkweed leaves dusted with drifting BT corn pollen are toxic to Monarch butterfly larvae, but some corn does not naturally pollinate at the same time of the year as butterflies are in their larval stage, so the effects may not be great. The full extent to which natural populations of butterflies are affected by BT corn is not known.

Herbicide-Resistant Crops

Some crops (e.g. corn) are being engineered to contain both herbicide tolerance and the BT toxin. Generally, the use of herbicide-tolerant crops will likely increase the use of herbicides. This has the potential to increase environmental pollution since it might increase the farmers' reliance on chemicals rather than mechanical and other means of weed control. SEE ALSO AGRICULTURAL BIOTECHNOLOGY; GENETICALLY MODIFIED FOODS; TRANS-GENIC PLANTS.

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Bioremediation

Bioremediation is the use of organisms to break down and thereby detoxify dangerous chemicals in the environment. Plants and microorganisms are used as bioremediators. The technology can take advantage of a natural metabolic pathway or genetically modify an organism to have a particular toxic "appetite."

Natural Microbial Bioremediators

On March 24, 1989, an oil tanker called the Exxon Valdez crashed into a reef in the Prince William Sound in Alaska, spilling 11 million gallons of oil that devastated the highly populated **ecosystem**. Attempts to clean rescued animals and scrub oily rocks were of little help and actually killed some organisms. Bioremediation was more successful. Ten weeks after the spill, researchers from the U.S. Environmental Protection Agency applied phosphorus and nitrogen fertilizers to 750 oil-soaked sites. The fertilizer stimulated the growth of natural populations of bacteria that metabolize polycyclic aromatic hydrocarbons, which are organic toxins that were present in the spilled oil. Over the next few years, ecologists monitored and compared the areas that the bacteria had colonized to areas where they did not grow, and found that the level of polycyclic aromatic hydrocarbons fell five times faster in the bioremediated areas.

Another environmental disaster being treated with natural bioremediation is the pollution of the Hudson River in New York with polychlorinated biphenyls (PCBs). General Electric Corporation deposited these compounds along a 40-mile stretch of the river between 1947 and 1977. PCBs were used to manufacture hydraulic fluids, capacitors, pigments, transformers, and electrical equipment. PCBs come in 209 different and interconverting forms, and the toxicity of a particular PCB depends upon the number of chlorine atoms it includes. Debate rages over whether it is better to remove and bury the most contaminated sediments, or to allow natural bacteria in the river to detoxify the PCBs.

The bioremediation of the Hudson River is occurring in three stages. First, buried **anaerobic** bacteria strip off chlorines. In the water column, aerobic bacteria cleave the two organic rings of the PCBs. Finally, other microorganisms degrade the dechlorinated, broken rings into carbon dioxide, water, and chloride. While the process effectively detoxifies the PCBs, it is a long-term process that can take up to two centuries.

ecosystem an ecological community and its environment

anaerobic without oxygen or not requiring oxygen



Cleanup crews use hot and cold high-pressure water jets to clean the rocks following the Exxon Valdez oil spill in 1994. This cleanup strategy was followed by the introduction of fertilizers to some areas of the spill, helping to encourage the growth of bacterial colonies.

vacuoles cell structures used for storage or related functions

polypeptides chains of amino acids

Natural Plant Bioremediators

For many millions of years, plants have adapted to the presence of various metals in varying amounts in soils. Some metals, such as zinc, nickel, cobalt, and copper, function as nutrients when eaten by humans in small amounts, but are toxic when consumed in excess. Heavy metals that are toxic even in trace amounts include mercury, lead, cadmium, silver, gold and chromium. Human activities such as mining, municipal waste disposal, and manufacturing have increased heavy metal pollution to dangerous levels in some areas. These chemicals cause oxidative damage, which destroys lipids, DNA, and proteins.

Certain plants, called hyperaccumulators, cope with excess heavy metals in the environment by taking them in and sequestering them in **vacuoles**, which are bubble-like structures in their cells. Sometimes the plant combines a pollutant with another molecule, a process called chelation. Organic acids often serve this role. Citric acid, for example, surrounds and thereby detoxifies cadmium, and malic acid does the same for zinc. A class of **polypeptides** called phytochelatins can also bind metals and escort them to vacuoles. Yet a third strategy that plants use to control metal accumulation is to employ a class of small, metal-binding proteins called metallothioneins. The intentional use of plants that use any of these ways to take heavy metals from soil is termed phytoremediation. It is a form of bioremediation.



This petroleum-contaminated dirt was mixed with clean dirt and manure before being injected with microbes. The microbes fed on the petroleum's hydrocarbons, converting the harmful materials into carbon dioxide and oxygen.

Natural phytoremediators can be amazing. Consider *Sebertia acuminata*, a tree that lives in the tropical rain forest of New Caledonia, near Australia. Up to 20 percent of the tree's dry weight is nickel. If slashed, the bark oozes a bright green. This plant can perhaps be used to clean up nickel-contaminated soil. Soybeans also preferentially take up nickel from soil. Another phytoremediator is *Astragalus*, also known as locoweed. It accumulates selenium from soil to counteract toxic effects of phosphorus, which tends to be abundant in selenium-rich soils. Cattle that munch on locoweed stagger about from selenium intoxication. Some plants act as sponges for metals in their environment. For example, plants that grow near gold mines assimilate gold into their tissues, apparently without harm. Prospectors use the gold content of such plants to locate deposits of the precious metal. Plants that grow near highways take up lead from gasoline exhaust. Near nuclear test sites, plants absorb radioactive strontium.

Genetically Modified Bioremediators

Biotechnology can transfer the ability to manufacture detoxifying proteins from one type of organism to another. One organism that was so modified has earned the distinction of being the first micro-organism to be patented. Called the "oil eater," the microorganism was actually a naturally occurring bacterium that had been given four **plasmids** that were also naturally occurring (plasmids are rings of DNA that can be transferred from one cell to another). It was the combination of the four transferred plasmids in a single bacterial cell that was novel and therefore patent-worthy. The four plasmids in the oil eater gave the bacterium the ability to degrade four components of crude oil. It was invented by Ananda Chakrabarty at General Electric in 1980.

Today, transgenic technology creates designer bioremediators. A transgenic organism contains a gene from another type of organism in all of its cells. The altered organism then manufactures the protein that the transgene encodes. The technology works because all organisms use the same genetic code. In other words, the same DNA and **RNA triplets** encode the same amino acids in all species.

plasmids small rings of DNA found in many bacteria

RNA triplets sets of three nucleotides

catalyzes aids in the reaction of

Transgenic bioremediation can engineer microbial metabolic reactions into plants whose root cells then produce the needed proteins and distribute them in the soil. For example, transgenic yellow poplar trees can thrive in soil that has been heavily contaminated with mercury if they have been given a bacterial gene that encodes the enzyme called mercuric reductase. This enzyme **catalyzes** the chemical reaction that converts a highly toxic form of mercury in soil to a less toxic gas. The leaves of the tree then emit the gas to the atmosphere, where it dissipates.

Cleaning up munitions dumps is yet another target of transgenic plants, with some interesting biological participants. In one approach, a bacterial gene that breaks down trinitrotoluene (TNT, the major component of dynamite and land mines) is linked to a jellyfish gene that makes the protein glow green. The bacteria can be spread directly on soil that is thought to contain weapons residues, or the genes can be transferred to various types of plants, whose roots then glow when they are near buried explosives. In the future, plants that have been genetically modified in several ways will be able to detect a variety of pollutants or toxins. SEE ALSO EUBACTERIA; TRANSGENIC ORGANISMS: ETHICAL ISSUES.

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Biotechnology

Biotechnology, broadly defined, refers to the manipulation of biology or a biological product for some human end. Before recorded history, humans grew selected plants for food and medicines. They bred animals for food, for work, and as pets. The ancient Egyptians learned how to maintain selected yeast cultures, which allowed them to bake and brew with predictable results. These are all examples of biotechnology. In more recent times, however, the term "biotechnology" has mainly been applied to specifically industrial processes that involve the use of biological systems. Today many biotechnology companies use processes that make use of genetically engineered microorganisms.

A Revolution in Biology

Following 1953, when Thomas Watson and Francis Crick published their famous paper on the double helix structure of DNA, a series of independent discoveries were made in chemistry, biochemistry, genetics, and microbiology, which together brought about a revolution in biology and led to the first experiments in genetic engineering in 1973. Because of this revolution, scientists learned to modify living microorganisms in a permanent, predictable way. Bacteria have been made to produce medical products, such as hormones, vaccines, and blood factors, that were formerly not available or available only

at great expense or in limited amounts. Crop plants have been developed with increased resistance to disease or insect pests, or with greater tolerance to frost or drought. What has made all these things possible is the collection of biochemical and molecular biological techniques for manipulating genes, which are the basic units of biological inheritance. These are the techniques used in genetic engineering or **recombinant DNA** technology.

The fusion of traditional industrial microbiology and genetic engineering in the late 1970s led to the development of the modern biotechnology industry. Using recombinant DNA technology, this industry has brought a long and steadily growing list of products into the marketplace. Human insulin produced by genetically engineered bacteria was one of the first of these products. It was followed by human growth hormone; an anti-viral protein called interferon; the immune stimulant called interleukin 2; a tissue plasminogen activator for dissolving blood clots; two blood-clotting factors, labeled VIII and IX, which are administered to hemophiliacs; and many other products.

Vitamin C

The production of certain chemicals has already become an important biotechnological industry. Vitamin C is a prime example. Humans, as well as other primates, guinea pigs, the Indian fruit bat, several species of fish, and a number of insects, all lack a key enzyme that is required to convert a sugar, glucose, into vitamin C.

No single bacterial genus or species is known that will carry out all of the reactions needed to synthesize vitamin C, but there are two (*Erwinia* species and *Corynebacterium* genus) that, between them, can perform all but one of the required steps. In 1985 a gene from one of these genus (*Corynebacterium*) was introduced into the second organism (*Erwinia herbicola*), resulting in a new bacterial form. This engineered organism can be used to produce a **precursor** to vitamin C that is converted via one chemical reaction into this essential vitamin. The engineering of many other microorganisms is being used to replace complex chemical reactions. For example, **amino acids**, needed for dietary supplements, are produced on a large scale using genetically modified microorganisms, as are antibiotics.

Laundry Detergents

Another important class of compounds produced by biotechnology is **enzymes**. These protein **catalysts** are used widely in both medical and industrial research. Proteases, enzymes that break down proteins, are particularly important in detergents, in tanning hides, in food processing, and in the chemical industry. One of the most significant commercial enzymes of this type is subtilisin, which is produced by a bacterium. Because many stains contain proteins, the manufacturers of laundry detergents include subtilisin in their product. Subtilisin is 274 amino acids long, and one of these, the methionine at position 222, lies right beside the active site of the enzyme. This is the site on the enzyme's surface where the substrate is bound, and where the reaction that is catalyzed by the enzyme takes place. In this instance the substrate is a protein in a stain, and the reaction results in the breaking of a peptide bond in the backbone of the protein. Unfortunately, methionine is an amino acid that is very easily **oxidized**, and laundry detergents are often used

recombinant DNA DNA formed by combining segments of DNA, usually from different types of organisms

Before Captain James Cook, the famous English sailor and navigator, had his men drink lime juice (which contains vitamin C) during extended sea voyages, many sailors fell ill or died of the vitamin C deficiency known as scurvy.

precursor a substance from which another is made

amino acids building blocks of protein

enzymes proteins that control a reaction in a cell

catalysts substances that speed up a reaction without being consumed

oxidized reacted with oxygen

in conjunction with bleach, which is a strong oxidizing agent. When used with bleach, the methionine in subtilisin is oxidized and the enzyme is inactivated, preventing the subtilisin from doing its work of breaking down the proteins present in food stains, blood stains, and the like.

To overcome this problem, genetic engineering techniques were used to isolate the gene for subtilisin, and the small part of the gene that codes for methionine 222 was replaced by chemically synthesized DNA fragments that coded for other amino acids. The experiment was done in such a way that nineteen new subtilisin genes were produced, and every possible amino acid was tried at position 222. Some of the altered genes gave rise to inactive versions of the enzyme, but others resulted in fully functional subtilisin. When these subtilisins were tested for their resistance to oxidation, most were found to be very good (except when cysteine replaced methionine: It too is easily oxidized). So now it is possible to use laundry detergent and bleach at the same time and still remove protein-based stains. This type of gene manipulation, which has been called “protein engineering,” has already been used for making beneficial changes in other industrial enzymes, and in proteins used for medical purposes.

Other Examples

Biotechnology companies are continuing to produce new products at an impressive rate. Numerous clinical testing procedures for human disorders such as AIDS and hepatitis and for disease-causing organisms such as those responsible for malaria and Legionnaires’ disease (a lung infection caused by the bacterium *Legionella pneumophila*), are based on diagnostic testing kits that have been developed by biotechnology companies. Many of these assays make use of recombinant **antibodies**, while others rely on DNA **primers** that are used in the polymerase chain reaction to detect DNA sequences present in an infecting organism, but not in the human genome.

Transgenic plants are now grown on millions of acres. Many of these plant species have been engineered to produce a protein, normally synthesized by the bacterium *Bacillus thuringiensis*, which is toxic to a number of agriculturally destructive insect pests but harmless to humans, most other non-insect animals, and many beneficial insects such as bees.

Ethical Issues

Like all industries, the biotechnology industry is subject to rules and regulations. Legal, social, and ethical concerns have been raised by the ability to genetically alter organisms. These have resulted in the establishment of governmental guidelines for the performance of biotechnology research, and specific requirements have been set to control the introduction of recombinant DNA products into the marketplace. General governmental guidelines for biotech research are published on the Internet at <http://www.aphis.usda.gov/biotech/OECD/usregs.htm>. Guidelines for plant genetic engineering and biotechnology are available at http://sbc.ucdavis.edu/Outreach/resource/US_gov.htm. SEE ALSO AGRICULTURAL BIOTECHNOLOGY; BIOTECHNOLOGY AND GENETIC ENGINEERING, HISTORY OF; CLONING GENES; CLONING ORGANISMS; GENE THERAPY; GENETIC TESTING; GENETICALLY MODIFIED FOODS; HEMOPHILIA; POLYMERASE CHAIN REACTION;

antibodies immune system proteins that bind to foreign molecules

primers short nucleotide sequences that help begin DNA replication

RECOMBINANT DNA; TRANSGENIC ANIMALS; TRANSGENIC MICROORGANISMS; TRANSGENIC PLANTS.

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Biotechnology Entrepreneur

The biotechnology entrepreneur often starts with a technical background, most commonly including scientific laboratory research. It is quite common for scientists who have spent a number of years in an academic environment to start a new company based on a technology platform or a novel discovery they made during their tenure at a university or research institute. Alternatively, a scientist or a group of scientists from a rather large company may decide to leave and start their own venture based on some of the research and development work they had either performed or thought about doing for their previous employer. This type of business may or may not be a spin-off from the parent company. In some cases, a businessperson or two may join the scientists in the start-up of the new company. A less frequent but still relatively common phenomenon is the start-up of a new venture by someone with a stronger background in business than in science. All of these individuals may receive financial support from venture capitalists, people who provide funding for new businesses in return for a share in future profits.

Whatever the background of the entrepreneur, there are a few key initial elements that are critical to achieving success. First and foremost is funding. There are a number of ways to secure funding, including small business research grants from the government, personal funds and loans, private "angel" investors, and venture capitalists. Entrepreneurs must become proficient at marketing themselves, their ideas, and their company in order to raise capital for the business. Equally important is having an attractive and viable business plan and business model, preferably based upon a novel technology or combination of technologies that has or can develop a strong patent position. It is also extremely important to have a good understanding of the market that will be served by the new business. Excellent communication skills are also crucial, as is the ability to recognize and recruit talented individuals, to hire them in the right order and place them in the positions that will best help the young company to focus, execute its plans, and grow.

A common mistake of the technically oriented entrepreneur is to try to completely manage both the technical and business side of the company

Business-minded scientists seek support from their peers at web sites such as <<http://www.bioe2e.org>>.

without consulting an experienced administrator to manage the business affairs. There are, however, the occasional entrepreneurs who can navigate both the business and technical sides with equal success. These leaders will usually retain the position of chief executive officer (CEO) as the company continues to grow. The technical entrepreneur who recruits experienced business people to help foster the growth of the company quite often ends up becoming the firm's chief scientific officer or the chief technical officer. This is probably the most commonly encountered scenario for companies that end up being traded on the stock exchange by releasing an initial public offering (IPO) of shares in the business.

A successful entrepreneur must also be an effective leader, able to contribute with creative ideas as well as to motivate staff and colleagues. He or she is often an individual with unending enthusiasm, a strong vision, and the ability to convince others that this vision will be successful. Entrepreneurs must have the conviction to do what is necessary to successfully execute the company's vision, while being flexible enough to adjust to new or different opportunities when they present themselves.

For a successful entrepreneur in the field of biotechnology, the rewards can be enormous. There is the satisfaction that comes from starting with an original idea and, through hard work, making it a reality, but there are financial rewards as well. Biotechnology entrepreneurs may draw initial salaries of \$150,000 or more, and potentially can claim very valuable stock options should their start-up companies eventually go public. A number of CEOs ultimately became multimillionaires on the strength of such options in the biotech boom of the late 1990s. SEE ALSO FINANCIAL ANALYST.

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Biotechnology: Ethical Issues

Biotechnology is the use of organisms or their parts or products to provide a valuable substance or process. **Fermentation** using microorganisms in brewing, baking, and cheese production are biotechnologies that date back centuries. Production of human insulin in bacteria to treat type I diabetes mellitus without causing allergic reactions is a more modern example of biotechnology. Two widely used biotechnologies that manipulate genes are recombinant DNA technology, which endows single-celled organisms with novel characteristics using genes from other organisms, and transgenic technology, which creates multicellular organisms that bear genes from other types of organisms. Genetically modified (GM) fruits and vegetables, such as a type of corn that manufactures a bacterial insecticide, are transgenic plants.

Ethical issues that arise from modern biotechnologies include the availability and use of privileged information, potential for ecological harm,

fermentation biochemical process of sugar breakdown without oxygen



An environmental activist opposed to the further development of genetically modified foods demonstrates outside Bangkok's UN headquarters during the opening session of the "New Technology Food and Crops: Science, Safety and Society" conference held July 10, 2001.



access to new drugs and treatments, and the idea of interfering with nature. Applications include agriculture and health care.

Agriculture

In agriculture, GM crops have been in the food supply in the United States for several years. Foods containing GM ingredients are not usually labeled to indicate their origin. This is because regulatory agencies determine food safety based on its similarity to existing foods, its chemical composition, and effects on the digestive systems of test animals, not on whether the plant variant arose from traditional agriculture or transgenic technology. If a food is found to include a chemical that could cause an allergy or is a toxin, it is

not marketed. As of early 2002, there have been no reports of harm coming from the consumption of GM foods.

Still, individuals who object to genetic modification would like the opportunity to select plant foods that were not produced in this manner. Labeling would solve this problem, and perhaps with continued consumer pressure it may come to pass. Some argue that at times, those who object to GM foods have acted in unethical ways. In several instances, protesters destroyed what they erroneously thought were fields of GM plants. Companies have behaved in ethically questionable ways in the GM food debate too. Before consumer outrage put an end to it, certain agrichemical companies sold GM crops that did not produce viable seed, forcing farmers to purchase new seed each year.

Another concern arising in agricultural biotechnology is the unintended spread of **transgenes** to other organisms. When a crop is grown in the field, its DNA, including the transgene, can theoretically be spread to other organisms in several ways. Certain types of plant viruses can transfer DNA from the host chromosome to a wild relative as well. Bacteria take up genes from the environment in the process known as transformation, and pass genes among different types of plants through **conjugation**. It is not yet known whether any of these latter processes have occurred with GM plant DNA, and detection may be difficult. It is likely to be a question not of whether but of when, however, given the large acreages devoted to GM plants.

Again the question arises of whether the consequences of such gene transfers are qualitatively different from the same process occurring on crop plants modified through traditional breeding. Opponents of GM crops say yes, since the potential exists to transfer genes from sources that would otherwise never be found in the agricultural environment. For instance, jellyfish genes are used in some agricultural research. In addition, the potential for harm from “escaped genes” may be greater precisely because the gene is so useful agriculturally. The gene for a natural insecticide may help grow safer corn, for example, but it could also allow a wild plant to escape its natural controls and become a serious forest weed. While such scenarios are hypothetical for the moment, opponents say that so little is known about the intricacies of ecology that caution is the only safe policy.

Health Care

In health care, genetic testing presents several ethical challenges. Legislation is in place or is being developed to limit access to genetic information, so that employers or insurers cannot discriminate against individuals because of their genotypes. Testing for a genetic disease presents a complication not seen in other types of illnesses, because the diagnosis of one individual immediately reveals the risk that other family members may be affected, based on the rules of inheritance. For example, a young woman learned that she is a carrier of Wiskott-Aldrich syndrome, which causes severe immune deficiency that is lethal in childhood. Because the mutant gene is carried on the X chromosome, each of her sons faces a 50 percent chance of inheriting the illness. Knowing that the same would be true for other carriers in her family, the young woman chose to inform all of her relatives who might also carry the gene. The decision of whether or not to be tested rests with the individuals.

transgenes genes introduced into an organism

conjugation a type of DNA exchange between bacteria

Another ethical dilemma in health care that arises from biotechnology is cost and access to new treatments. Such drugs as tissue plasminogen activator, used to break up clots that cause heart attacks and strokes, and erythropoietin and colony-stimulating factors, used to restore blood supplies in cancer patients being treated with chemotherapy, are extremely expensive. Although insurers often cover the costs in the United States, people in many other nations cannot take advantage of these drugs.

Because biotechnology is a rapidly evolving field, experiments and **clinical trials** are ongoing. Participation in a clinical trial of a recombinant DNA-derived drug, or of a gene therapy, requires informed consent. A case in 1999 provoked reevaluation of the care with which such participants are screened, and of the adequacy of informed consent protocols. Jesse Gelsinger was 19 when he received experimental gene therapy to treat ornithine transcarbamylase deficiency. In this disorder, lack of an enzyme that metabolizes protein leads to buildup of ammonia, which damages the brain. Most affected individuals die within days of birth, but survivors can usually control symptoms with diet and drugs, as Jesse had been doing. This is why his death five days after receiving the gene therapy was especially tragic. An underlying and undetected medical condition may have contributed to his death.

Medication usually controlled Gelsinger's condition, but he chose to join the clinical trial so that he might help babies who died of a more severe form of the illness. Although Gelsinger clearly stated that he realized he might die, questions arose about the extent of his knowledge, largely because he had been healthy. This issue does not arise in the more common situation in which a participant in a gene therapy trial has exhausted conventional treatments, because he or she has little to lose at that point. Clinical trials of gene therapy are now conducted with much greater care.

clinical trials tests performed on human subjects

New Challenges

Another objection to biotechnology is that it interferes with nature, but so do traditional agriculture and medicine. However, the changes that biotechnology can introduce are usually quite unlikely to occur naturally, such as a tobacco plant that glows thanks to a firefly protein, or cloning a human. We place limits on some biotechnologies, but not on others, based on our perceptions and on the intents of the interventions. The glowing tobacco plant was done as an experiment to see if a plant could express a gene from an animal, but many countries ban human cloning because it is seen as unnecessary, dangerous, and unethical. Still, time can change minds. When Louise Joy Brown, the first baby conceived using in vitro fertilization, was born in 1980, objection to "test tube baby" technology was loud. The procedure is now routine. In general, it seems that a biotechnology will eventually be considered ethical if evidence accumulates demonstrating that it does no harm.

A biotechnology that by its very definition causes harm is bioterrorism, especially when genetic manipulation is used to augment the killing power of a naturally occurring pathogen. Bioterrorism dates back to the Middle Ages, when Tartan warriors hurled plague-ridden corpses over city walls to kill the inhabitants. The British used a similar approach in the eighteenth century, when they intentionally gave Native Americans blankets that carried smallpox virus. Efforts in the former Soviet Union to create bio-weapons

